Min Number Of Coins For Change

AlgoExpert · Medium · Dynamic Programming

The problem

- Fewest coins summing to n, or -1 if impossible.

Intuition

- Build the fewest coins for every amount from zero up to n.

Approach

- 1. Array of infinities, zero for amount zero
- 2. For each coin, relax amounts from the coin up
- 3. Answer or minus one

Java solution

```java
public class MinNumberOfCoinsForChange {
    public static int minNumberOfCoinsForChange(int n, int[] denoms) {
        int[] dp = new int[n + 1];
        Arrays.fill(dp, Integer.MAX_VALUE);
        dp[0] = 0;
        for (int d : denoms)
            for (int a = d; a <= n; a++)
                if (dp[a - d] != Integer.MAX_VALUE) dp[a] = Math.min(dp[a], dp[a - d] + 1);
        return dp[n] == Integer.MAX_VALUE ? -1 : dp[n];
    }
}
```

Complexity

- Time: O(n*d)
- Space: O(n)

Recap

- Build the fewest coins for every amount from zero up to n.
- Complexity: O(n*d)